

Scalability & Future Plan

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how Green-Predict-Pro can scale to handle larger user bases, increased data loads and extended features, along with the team's future roadmap.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Hosted on free-tier cloud with limited concurrent users.	Slow response during peak load.	High
2	Only English UI available.	Limits adoption by non-English-speaking farmers.	Medium
3	Model trained on a single public dataset.	May not fully reflect regional variations.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	~100 concurrent users on free tier.	Move to Gunicorn + Nginx on a paid Render / AWS EC2 instance; add horizontal scaling.
2	Data Storage	CSV dataset + SQLite.	Migrate to MySQL / PostgreSQL and store dataset in cloud object storage.
3	Performance	Single-process Flask.	Async workers, model caching in memory, CDN for static assets.
4	Security	Basic Flask-Login auth.	Add JWT, rate limiting, WAF and periodic security audits.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Weather API auto-fill for temperature, humidity and rainfall.	Sep 2026	Reduces manual entry and improves accuracy.
Phase 3	Multilingual UI (Telugu, Hindi, Tamil, Kannada).	Dec 2026	Widens adoption across Indian states.
Phase 4	Mobile app + satellite soil-moisture integration + fertilizer advisory.	2027	Complete precision-agriculture assistant.